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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	奶龙
Selected Domain	Domain2
Selected SDG Track (1–17)	8.2 8.3 10 12
Project / Solution Name	Nexus
Name (Leader)	LING YUXUAN
Email & Phone (Leader)	1730211684@qq.com 1123232813
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Many small and medium-sized enterprises still use notebooks for bookkeeping, paper documents for receipts, and month-end check-in-one transactions, relying only on basic spreadsheets for financial management. Reports say this is time-consuming, prone to errors, and lacks real-time data, making it difficult for companies to digitize large amounts of data and unable to afford customized systems.

2.2 The Proposed AI Solution

[Maximum 300 words]

My project emphasizes integration, using AI to integrate all aspects of work, greatly reducing the cost for enterprises to consolidate data, enabling real-time and accurate data presentation. We also offer various customization solutions tailored to different types of enterprises, solving the pain points that small and medium-sized enterprises cannot afford customization. Secondly, my software is open-ended, allowing clients to import various AI, spreadsheets, and other data, with built-in AI helping you organize your data

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: HTML5, CSS3 and native JavaScript ES Modules, with responsive layouts. Business logic: a JavaScript rules engine checks inventory and staff availability, demonstrates role permissions, supports human approvals, prevents duplicate operations and creates tasks. Business records remain in browser localStorage without cross-device synchronization. Backend: a Sites-hosted Cloudflare Worker authenticates AI calls through ChatGPT sign-in; Cloudflare D1 stores daily AI usage counters only. The OpenAI Responses API (gpt-4.1-mini) extracts requirements and drafts replies for human review; live generation is currently blocked by insufficient API credits. Local Excel/CSV import uses fflate 0.8.3. Other vendor integrations remain proposed OpenAPI contracts. Node.js, GitHub and GitHub Actions support tests/builds. OpenAI Sites hosts the prototype. SDC-provided Replit premium access supported development/testing.

Customer messages → Customer service confirmation of requirements → Business rule checks → Manager approvals → Creating employee tasks → Activity recording and tracking. Customers send messages to customer service through an independent entry point, which organizes and confirms requirements such as quantity and deadlines. The rules engine reads inventory, employee skills, approved leave, and project data, checks for stockouts and personnel conflicts, and proposes assignment suggestions. After manager confirmation, the system re-checks business information, creates tasks for corresponding employees, and records approvals, exceptions, and handling results

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We adopt a design mindset of "understanding the problem—prototyping—testing iterations," focusing on the burdens of small businesses repeatedly following up on orders, verifying information, and assigning tasks. Nexus shifts customer requirements into business checks, manual approvals, and employee tasks, clearly demonstrating operations by role to reduce learning costs. Core rules already support automated testing. Before the second phase, we plan to invite target users to test, simplify the process based on feedback, and supplement authentication, database, and authorized API connections. After permissions, exception recovery, and end-to-end testing, we will conduct a small-scale pilot

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Mainly corresponding to SDG 8.2: improving economic productivity through technological upgrades and innovation; while also supporting SDG 8.3 for the development of micro, small, and medium-sized enterprises SDG source: https://sdgs.un.org/goals/goal8
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Estimated manual baseline: 10 minutes per request (3 minutes to structure requirements, 4 to check resources, 3 to assign and record). This is a team estimate, not measured industry data; it will be validated with target users.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median active handling time per request (minutes); percentage change against the manual baseline; percentage of critical actions requiring human approval. Track missing information and failures.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Reduce the manual processing time for each request by 30%, from the estimated 10 minutes to under 7 minutes; Achieve 100% manual approval coverage for critical operations.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
fflate 0.8.3	Code library	MIT
Node.js	Runtime / tooling	MIT; bundled notices
GPT-4.1 mini	Pre-trained API model	OpenAI API terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]LING YUXUAN LI JUNZHANG XIE WEIYE HU ZHAOMING

[Date] 24/10/2026